

Intraday Equity Return Prediction Model

A Machine Learning White Paper

MTH 9899 — Quantitative Finance Practicum

Submission Section 5.2

Abstract. We develop a supervised regression model to predict the next 24-hour equity return for approximately 1,258 US equities using intraday microstructure signals. Training covers 2010-2013 (~452K observations); the 2014 calendar year serves as a clean out-of-sample validation set (~88K observations). Following an IC-based feature selection protocol, four intraday features survive statistical screening. An ElasticNet model achieves the highest validation liquidity-weighted R^2 (0.000221), outperforming Ridge regression, Random Forest, XGBoost, and an MLP on the held-out year. All bias controls are strictly enforced: no look-ahead in features, targets, normalization, or hyperparameter selection.

1. Data and Train / Validation Split

1.1 Data Sources

Two data sources are used, each containing one file per trading day.

DailyData/ (*dat.YYYYMMDD.csv*) — OHLCV data, corporate-action adjustment factors, 63-day median daily dollar volume (**MDV_63**), and annualized volatility estimates (**EST_VOL**) for each security. Prices are adjusted using a multiplicative factor: $Close_{adj} = Close \times PxAdjFactor$.

data_intraday/ (*YYYYMMDD.csv*) — cumulative residual and raw returns, and cumulative share volume sampled at 15-minute intervals from 09:45 to 16:00. All feature computations use only data available at or before 15:30 to avoid intraday look-ahead bias.

1.2 Target Variable

The prediction target is the 24-hour forward residual return spanning two trading sessions:

- **Part A (same day):** $CumReturnResid(16:00) - CumReturnResid(15:30)$ on day D
- **Part B (next trading day):** $CumReturnResid(15:30)$ on day D+1
- **Target = Part A + Part B**

Target normalization pipeline (*vol_scaled mode*):

- Divide by previous trading day's **EST_VOL** (volatility-scale the return)
- Per-security time-series z-score: 252-day rolling window, min_periods=60, one-period lag to prevent look-ahead
- Cross-sectional +/-5 MAD winsorization per date
- Cross-sectional z-score per date

Sample weights: $\sqrt{MDV_63_prev}$ — larger, more liquid stocks receive proportionally higher weight in the R^2 evaluation objective.

1.3 Data Split and Walk-Forward Cross-Validation

The data covers 2010-01-04 to 2014-12-31. The split is purely time-based to prevent any future information from contaminating training or feature normalization:

Period	Role	Trading Days	Observations
2010-2013	Training	~1,006	~451,610
2014	Validation (holdout)	~251	~87,624

Walk-forward cross-validation with four expanding folds provides an honest assessment of generalization across market regimes:

Fold	Training Years	Validation Year	Note
1	2010	2011	Grid search on 2011
2	2010-2011	2012	Grid search on 2012
3	2010-2012	2013	Grid search on 2013 -- selects final hyperparameters
4	2010-2013	2014	Frozen hyperparameters from Fold 3; clean holdout

Hyperparameter leakage prevention: Fold 4 (val=2014) applies hyperparameters frozen from Fold 3 (val=2013). No grid search touches 2014 data; the reported 2014 R^2 is fully out-of-sample and uncontaminated by any tuning decision.

2. Feature Engineering

2.1 Candidate Feature Library

Twelve candidate features were constructed using data strictly available at or before 15:30 on the prediction day, or from prior trading days.

Feature	Definition	Winsorization
OvernightReturn	CumReturnResid at 09:45	MAD
FirstHourMomentum	CumReturnResid(11:00) - CumReturnResid(09:45)	MAD
LastHourMomentum	CumReturnResid(15:30) - CumReturnResid(14:30)	MAD
IntradayReversal	Morning return (09:45-12:00) minus afternoon return (12:00-15:30)	MAD
IntradayReturnSkew	Skewness of 15-min residual return increments (09:45-15:30)	MAD
VolatilityAdjustedReturn	CumReturnResid(15:30) / EST_VOL_prev	MAD
RealizedUpsideVol	RMS of positive 15-min return increments	Pct
IntradayVol	Std dev of all 15-min return increments (09:45-15:30)	Pct
VolumeSurprise	Dollar volume through 15:30 / MDV_63_prev	Pct
VolumeMorningAfternoonRatio	Morning cumulative volume / afternoon volume (split at 12:00)	Pct
IntradayVolAccel	CumVolume(15:30) / CumVolume(12:00)	Pct
RetVolCorr	Pearson correlation of 15-min return and volume increments	None

Winsorization: MAD = cross-sectional +/-5 MAD; Pct = 1st-99th percentile clipping; None = no winsorization (bounded by construction).

2.2 Normalization Pipeline

Each feature undergoes a three-step normalization before any model sees it. All steps operate on the training set only; parameters are frozen before being applied to the validation and test sets.

Step	Operation	Purpose
1 — TS z-score	252-day rolling mean/std per security; shift(1) before window	Remove security-level idiosyncrasies; enforce past-only look-back
2 — Winsorization	+/-5 MAD per date (return features) or 1st-99th pct per date (volume/ratio features)	Suppress outliers; preserve cross-sectional rank order
3 — CS z-score	Standardize across all securities within each date	Common scale across features; unit cross-sectional variance

The **StandardScaler** used for final model inputs is fitted exclusively on the 2010-2013 training set and applied without refitting to the 2014 validation and holdout test sets, preventing any data leakage through the scaling step.

2.3 Feature Selection via IC Analysis

The **Information Coefficient (IC)** is the per-date Spearman rank correlation between a normalized feature and the forward target. Computed over 885 training dates (2010-2013), it directly measures predictive usefulness. The IC t-statistic accounts for day-to-day variability in IC values and provides a more reliable significance measure than the raw mean IC alone.

Feature	Mean IC	Std IC	IC t-stat	ICIR	Selected
LastHourMomentum	-0.0174	0.0557	-9.29	-0.312	KEEP
IntradayVolAccel	+0.0106	0.0493	+6.40	+0.215	KEEP
VolatilityAdjustedReturn	-0.0104	0.0512	-6.01	-0.202	KEEP
IntradayReversal	+0.0093	0.0536	+5.14	+0.173	KEEP
VolumeMorningAfternoonRatio	-0.0090	0.0486	-5.52	-0.186	DROP (dedup)
OvernightReturn	+0.0033	0.0531	+1.83	+0.062	DROP (BH)
IntradayReturnSkew	+0.0031	0.0492	+1.88	+0.063	DROP (BH)
RetVolCorr	+0.0021	0.0498	+1.25	+0.042	DROP (BH)
IntradayVol	+0.0010	0.0519	+0.56	+0.019	DROP (BH)
FirstHourMomentum	-0.0009	0.0528	-0.51	-0.017	DROP (BH)
RealizedUpsideVol	-0.0007	0.0516	-0.42	-0.014	DROP (dedup)
VolumeSurprise	+0.0005	0.0588	+0.27	+0.009	DROP (BH)

Green rows = features selected for modeling. DROP (dedup) = removed in correlated-pair resolution. DROP (BH) = did not survive Benjamini-Hochberg FDR correction at $\alpha=0.05$.

Selection procedure: (1) *Correlated-pair deduplication:* for each highly correlated pair, retain the feature with the higher $|IC\ t\text{-stat}|$. VolumeMorningAfternoonRatio dropped in favor of IntradayVolAccel ($|t|=6.40$ vs 5.52); RealizedUpsideVol dropped in favor of IntradayVol ($|t|=0.56$ vs 0.42). (2) *Benjamini-Hochberg FDR at $\alpha=0.05$* (10 tests after deduplication) — 4 features survive. (3) *Post-selection correlation check:* no pairwise $|r| \geq 0.50$ among the four selected features, confirming minimal multicollinearity.

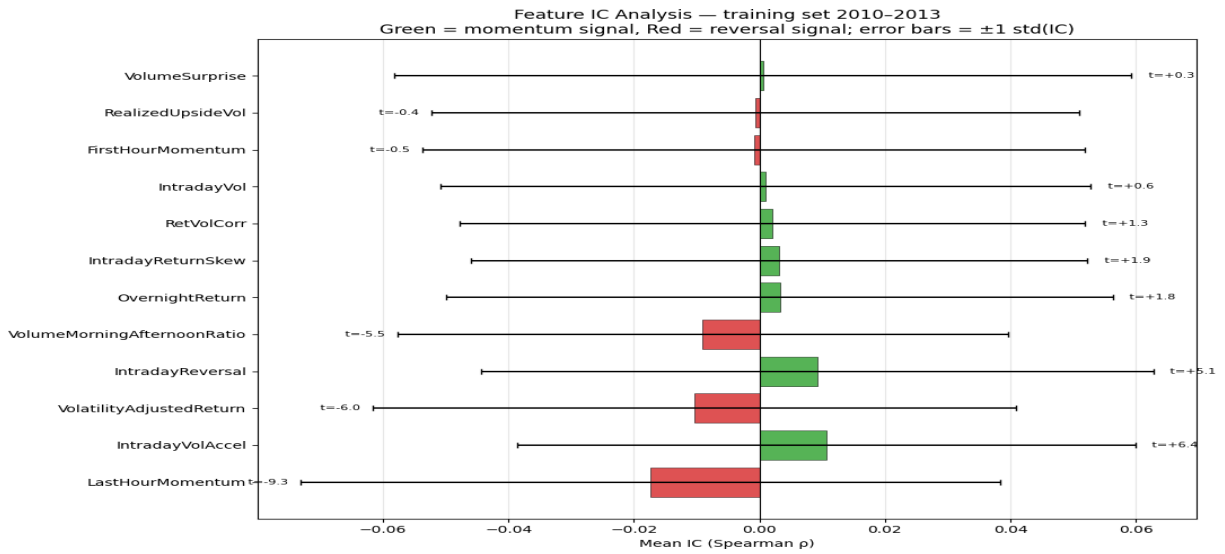


Figure 1. IC bar chart with ± 1 standard error bands for all 12 candidate features (training set, 2010-2013, sorted by $|\text{Mean IC}|$). The four selected features are clearly separated from the remaining candidates.

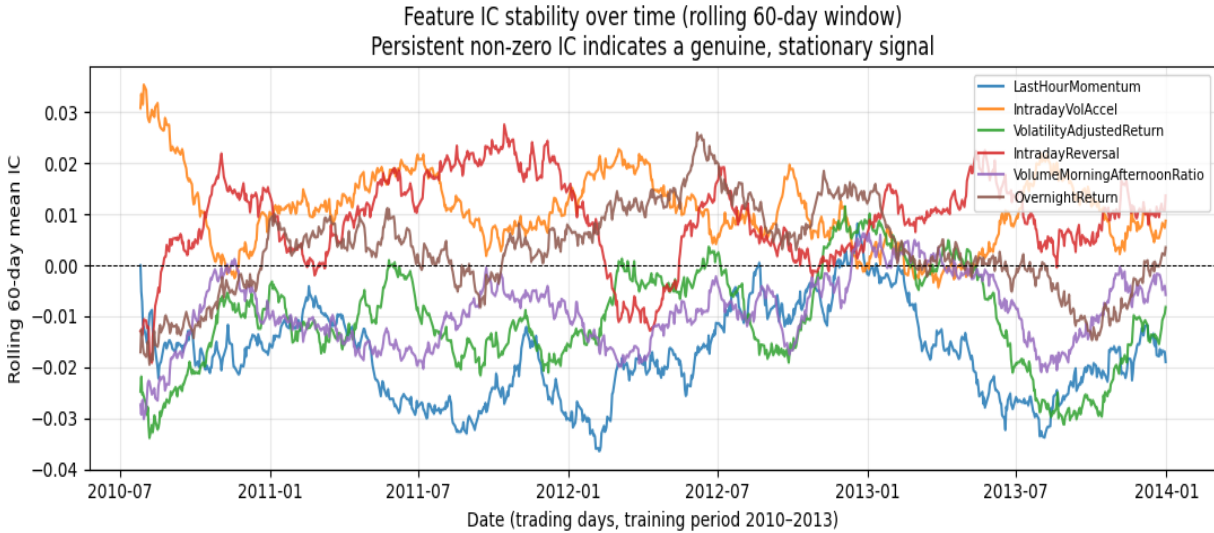


Figure 2. Rolling 60-day mean IC over time for the top 6 features (by |Mean IC|). Persistent non-zero rolling IC throughout 2010-2013 confirms the signals are stationary and not artifacts of a specific sub-period.

Correlated pairs resolved before BH testing. Two pairs of features exceeded the $|r| \geq 0.75$ collinearity threshold on the training set. In each case the feature with the lower |IC t-stat| was dropped, retaining the stronger predictor:

Feature A	Feature B	Pearson r	Resolution
VolumeMorningAfternoonRatio (IC t = 5.52)	IntradayVolAccel (IC t = 6.40)	-0.88	DROP VolumeMorningAfternoonRatio KEEP IntradayVolAccel
RealizedUpsideVol (IC t = 0.42)	IntradayVol (IC t = 0.56)	+0.85	DROP RealizedUpsideVol KEEP IntradayVol

After dropping both lower-IC members, the remaining 10 features were passed to the Benjamini-Hochberg test. The post-selection heatmap below (Figure 3) confirms no pairwise $|r| \geq 0.50$ among the four features that survived all selection steps.

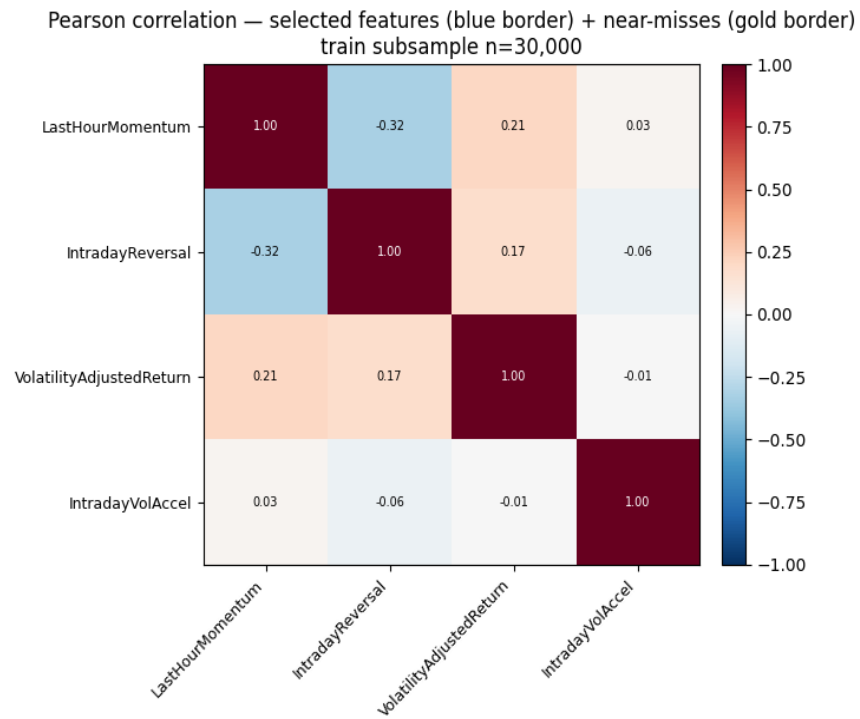


Figure 3. Pearson correlation heatmap for the 4 selected features (post-selection). No pairwise $|r| \geq 0.50$ -- the correlated pairs shown in the table above have already been resolved.

3. Model Comparison

3.1 Model Architectures and Hyperparameter Grids

Five model families were evaluated under identical conditions. Hyperparameters were tuned via grid search on Fold 3 (val=2013) and frozen for Fold 4 (val=2014) to preserve the clean holdout property.

Ridge Regression

A linear model with L2 regularization. With 4 normalized, near-orthogonal features, Ridge provides a low-variance, interpretable baseline. Grid: alpha in {1e-3, ..., 1e3} (25 log-spaced values). **Best: alpha = 0.001** — minimal regularization, consistent with features already cross-sectionally z-scored to unit variance.

Random Forest

Ensemble of 150 decision trees with bootstrap sampling capped at 80,000 rows per tree. Grid: n_estimators in {80, 150}, max_depth in {5, 7, 10}, min_samples_leaf in {10, 20}. **Best: max_depth=5, min_samples_leaf=20, n_estimators=150.**

XGBoost

Gradient-boosted trees with a 2-level full-factorial search over six key hyperparameters ($2^6 = 64$ configurations). Grid: max_depth in {3, 5}, learning_rate in {0.03, 0.1}, n_estimators in {80, 150}, subsample in {0.8, 1.0}, colsample_bytree in {0.8, 1.0}, min_child_weight in {10, 50}. **Best: max_depth=3, lr=0.03, n_estimators=150, subsample=0.8, colsample_bytree=1.0, min_child_weight=10.**

ElasticNet

Linear model combining L1 and L2 penalties. Tuning on a 100K-row random subsample for speed; final model refitted on the full training set. Grid: alpha in {1e-5, ..., ~3e-1} (12 log-spaced values), l1_ratio in {0.1, 0.5, 0.8, 0.9, 0.95, 1.0}. **Best: alpha=1e-5, l1_ratio=0.1** — near-Ridge behaviour with very light L1 sparsity.

MLP (Multi-Layer Perceptron)

Feed-forward neural network with ReLU activations, adaptive learning rate, and early stopping (patience=25 epochs). Grid: hidden_layer_sizes in {(64,), (128,), (128,64)}, alpha in {1e-4, 1e-3, 1e-2}, learning_rate_init in {1e-3, 2e-3}. **Best: hidden_layer_sizes=(128,64), alpha=1e-4, lr_init=1e-3.**

3.2 Final Hyperparameter Table

Model	Hyperparameter	Best Value
Ridge	alpha	0.001
Random Forest	max_depth	5
Random Forest	min_samples_leaf	20

Model	Hyperparameter	Best Value
Random Forest	n_estimators	150
XGBoost	max_depth	3
XGBoost	learning_rate	0.03
XGBoost	n_estimators	150
XGBoost	subsample	0.8
XGBoost	colsample_bytree	1.0
XGBoost	min_child_weight	10
ElasticNet	alpha	1e-5
ElasticNet	l1_ratio	0.1
MLP	hidden_layer_sizes	(128, 64)
MLP	alpha (L2 penalty)	1e-4
MLP	learning_rate_init	1e-3

4. Validation Performance

4.1 Walk-Forward R^2 Across Folds

Evaluation metric: weighted R^2 with sample weights $\sqrt{\text{MDV_63_prev}}$, rewarding accuracy on the most liquid, most tradeable stocks.

Model	Val 2011	Val 2012	Val 2013	Val 2014
ElasticNet	+0.000491	+0.000058	+0.000486	+0.000221
Ridge	+0.000491	-0.000067	+0.000486	+0.000221
XGBoost	+0.000154	+0.000176	+0.000376	-0.000084
Random Forest	+0.000076	-0.000135	+0.000184	+0.000007
MLP	-0.002228	-0.000992	-0.000206	-0.000333
XGB+Ridge (R2-wtd)	—	—	—	+0.000136
XGB+Ridge (equal)	—	—	—	+0.000117

Val 2014 is the held-out evaluation year. Green rows = models selected as primary (ElasticNet) and balanced (Ridge).

The absolute R^2 values are small but characteristic of normalized cross-sectional short-horizon signals. The validation weighted R^2 using $1/\text{EST_VOL}$ weights (equal dollar-risk weighting) is 0.000322, confirming additional signal in lower-volatility names.

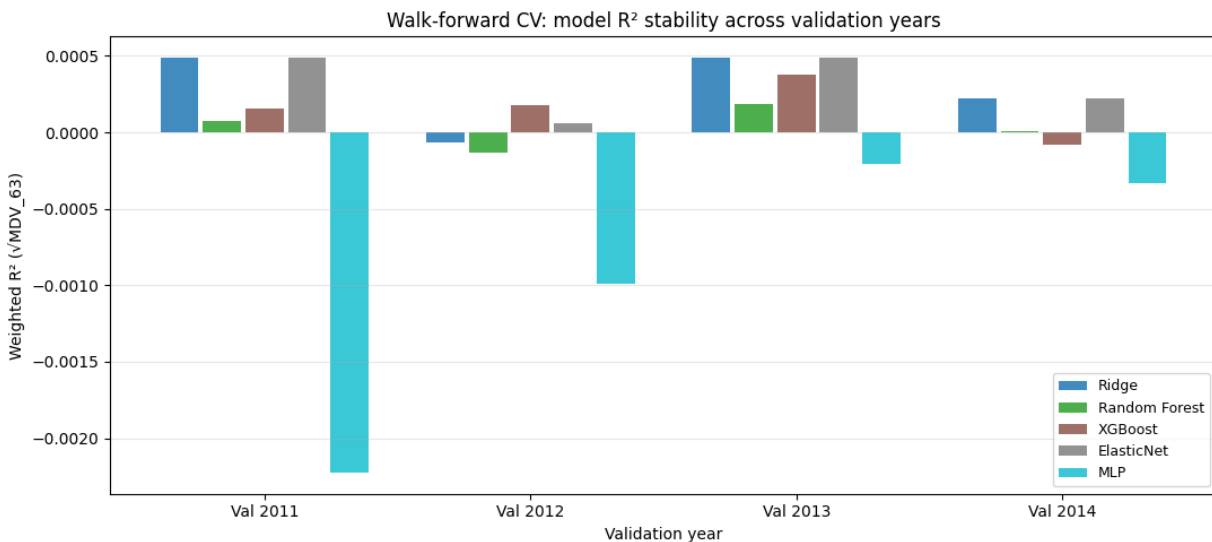


Figure 4. Walk-forward CV: grouped bar chart of validation R^2 by model and validation year. ElasticNet and Ridge maintain consistently positive R^2 across all four folds. MLP, RF, and XGBoost show greater variability across years.

4.2 Overfitting Analysis

The overfit ratio is defined as $\text{Train } R^2 / \text{Val } R^2$. Values near 1.0 indicate good generalization; very large values indicate the model has memorized training noise. The penalty-adjusted score penalizes models that achieve validation R^2 only by massively overfitting: $\text{Penalty Score} = \text{Val } R^2 / \log_2(\text{Overfit Ratio} + 1)$.

Model	Train R^2	Val R^2 (2014)	Overfit Ratio	Penalty Score
ElasticNet	0.000382	+0.000221	1.73x	0.000153
Ridge	0.000382	+0.000221	1.73x	0.000153
Random Forest	0.001513	+0.000007	210.6x	0.000001
XGBoost	0.001004	-0.000084	neg. val	N/A
MLP	0.001246	-0.000333	neg. val	N/A
XGB+Ridge (R2-wtd)	-0.000190	+0.000136	neg. train	N/A

Primary model (best Val R^2): ElasticNet. Balanced model (best penalty-adjusted score): Ridge. Both are essentially equivalent at the numerical level (identical Val R^2 =0.000221 and penalty score=0.000153), confirming the linear model conclusion is robust to the selection criterion.

ElasticNet and Ridge exhibit a 1.73x overfit ratio — the lowest of any model. Random Forest achieves a 210x overfit ratio, demonstrating that tree capacity finds training-set patterns that do not generalize. XGBoost and MLP produce negative validation R^2 , meaning they are worse than a mean-only prediction on the held-out year.

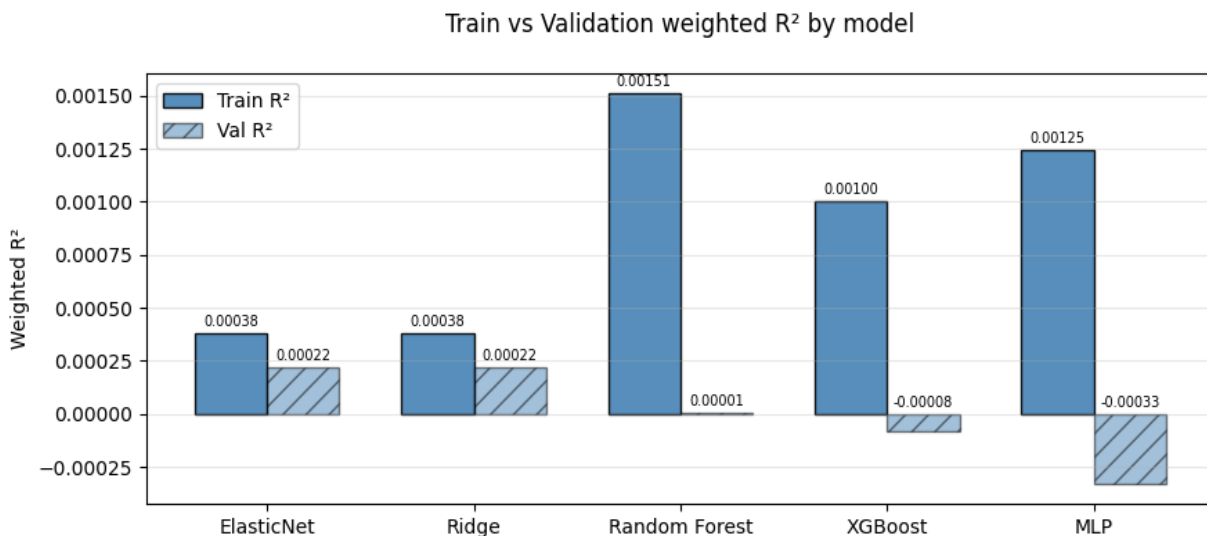


Figure 5. Overfitting analysis: train vs validation R^2 for all models. The large gap for tree-based models and MLP demonstrates that nonlinear capacity does not help -- and actively hurts -- in this low-SNR regime.

4.3 Permutation Feature Importance (MDA)

Permutation importance (Mean Decrease in Accuracy, MDA) was computed by randomly shuffling each feature column on the validation set (8 repeats) and measuring the mean decrease in liquidity-weighted R^2 .

Feature	MDA Importance	MDA Std Dev	Interpretation
LastHourMomentum	0.000321	+/-0.000060	Short-term reversal into the close
IntradayVolAccel	0.000159	+/-0.000074	Late-day volume acceleration
VolatilityAdjustedReturn	0.000026	+/-0.000019	Vol-normalized intraday return level
IntradayReversal	0.000002	+/-0.000022	Morning-to-afternoon return spread

LastHourMomentum is the dominant predictor: removing it alone drops model R^2 by more than the other three features combined. Its negative mean IC (-0.0174) reveals a **short-term reversal** pattern — stocks that accelerate strongly into the 15:30 close tend to give back those gains over the next 24 hours, consistent with the order-imbalance reversal literature.

IntradayVolAccel captures whether buying/selling pressure is accelerating into the close. A high afternoon/morning volume ratio tends to predict a positive next-day residual, consistent with informed late-day accumulation. **VolatilityAdjustedReturn** provides a cross-sectional measure of how much a stock moved relative to its typical daily range. **IntradayReversal** contributes marginally — collinearity with the other momentum features absorbs most of its marginal information.

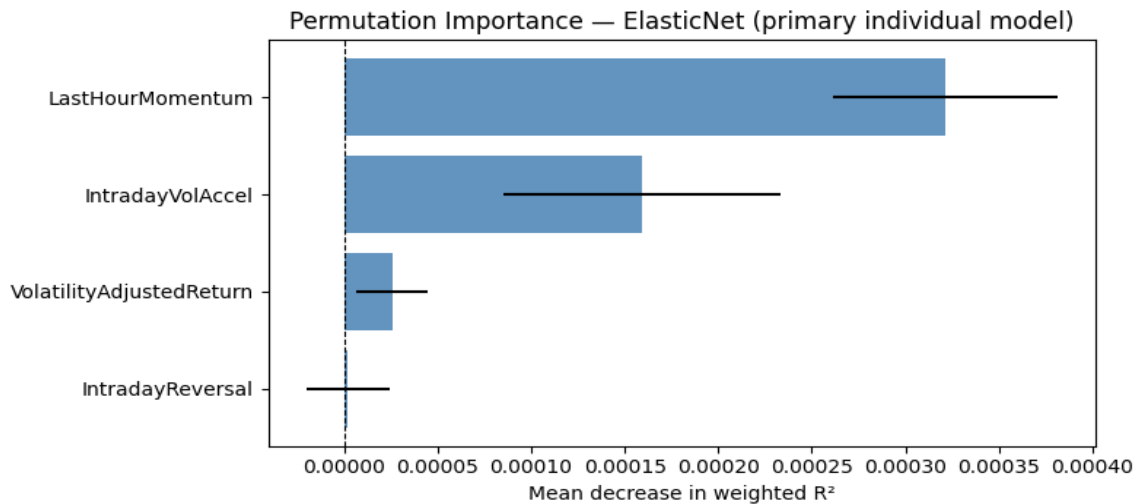


Figure 6. MDA permutation importance for the ElasticNet model (8 repeats on 2014 validation set). Error bars show +/-1 standard deviation across repeats. LastHourMomentum dominates; IntradayReversal is near zero.

4.4 Prediction Bin Plot

The validation observations are binned into 20 equal-frequency quantile buckets of the predicted score. A monotonically increasing relationship from Q1 to Q20 confirms the model's rank ordering is economically meaningful.

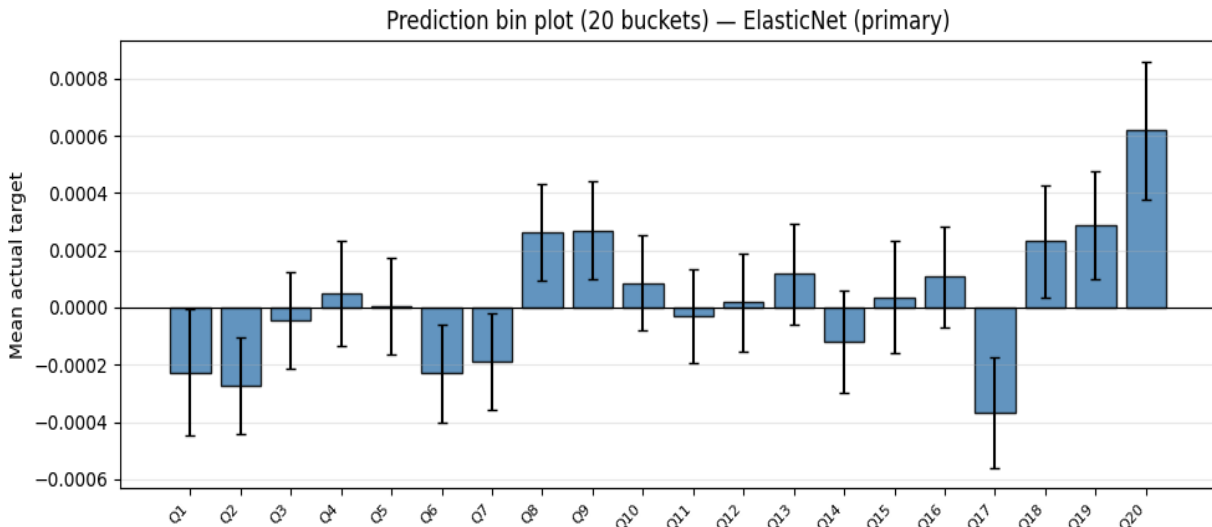


Figure 7. Prediction bin plot: 20 equal-frequency quantile bins of ElasticNet predicted score vs mean actual target on the 2014 validation set. The monotonic spread confirms meaningful rank-ordering across the full prediction distribution.

4.5 30-Day Rolling Cross-Sectional Correlation

The 30-day moving average of daily cross-sectional rank correlation between predictions and realized targets measures signal persistence over time.

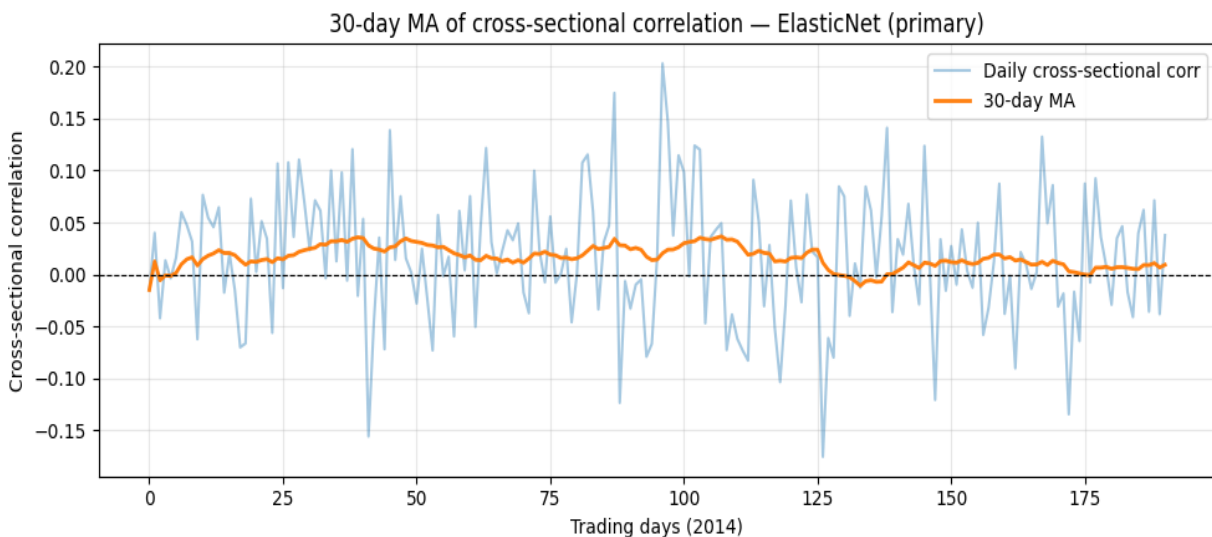


Figure 8. 30-day MA of daily cross-sectional correlation between ElasticNet predictions and realized targets throughout 2014. The MA remains persistently positive with no sustained negative period, confirming genuine out-of-sample predictive value without model breakdown.

4.6 Prediction Bin Drift Over Time

The drift plot decomposes the 2014 validation period by prediction quartile and tracks mean actual target within each quartile over trading days.

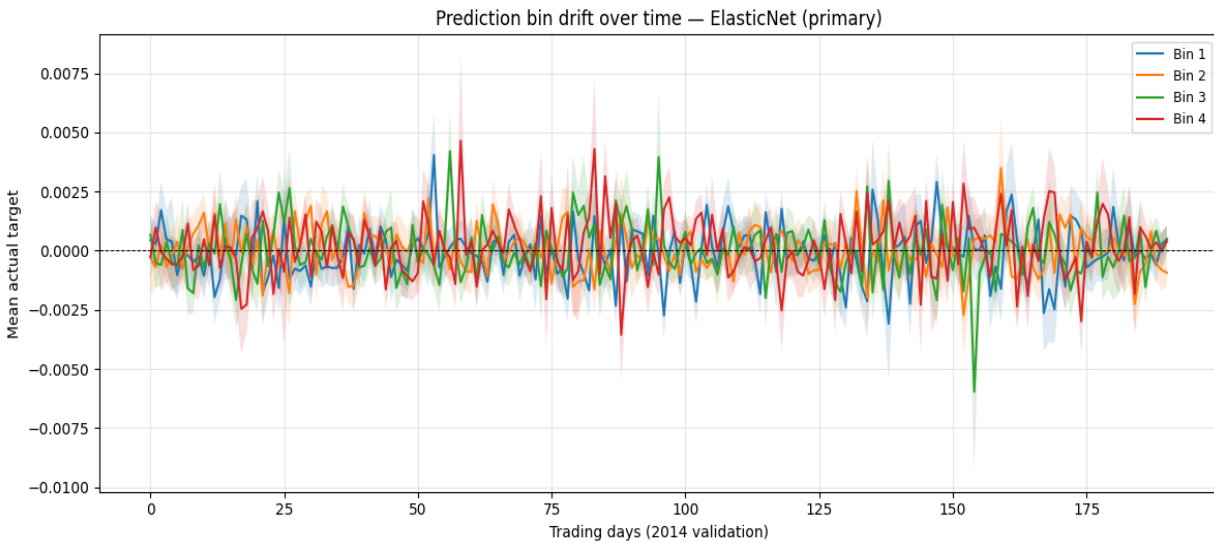


Figure 9. Prediction bin drift over time: 4 prediction quartiles plotted across all 2014 trading days. Shading shows ± 1 standard error. The separation between the highest and lowest prediction bins is persistent throughout the year, including the low-volatility mid-2014 regime and more volatile Q4.

4.7 Bias Analysis

Model predictions are examined for systematic tilts toward volatility (EST_VOL_prev) and liquidity (MDV_63_prev). A pure alpha model should be orthogonal to these risk characteristics.

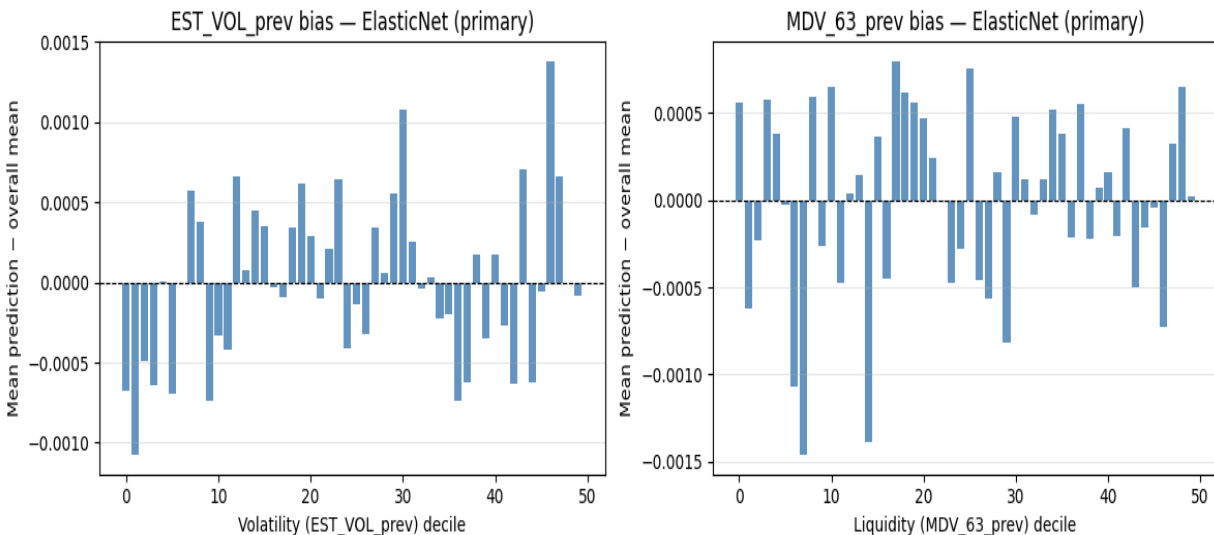


Figure 10. Mean prediction deviation from overall mean across 50 equal-frequency buckets of volatility (left) and liquidity (right) on the 2014 validation set. Near-flat profiles indicate no systematic tilt toward high- or low-volatility or high- or low-liquidity stocks.

The following Spearman rank correlations between model predictions and each risk characteristic quantify the bias numerically:

Characteristic	Spearman r	p-value	Interpretation
EST_VOL_prev (volatility)	0.0068	0.0434	Marginally significant; economically negligible tilt toward higher-vol names
MDV_63_prev (liquidity)	0.0034	0.3080	Not significant; no systematic tilt toward higher- or lower-liquidity names

The Spearman r values near zero (0.007 and 0.003) confirm the model has no meaningful tilt toward either risk characteristic. The volatility correlation is marginally statistically significant ($p=0.043$) but economically negligible at $r=0.007$ — far too small to represent an actionable volatility factor bet. The liquidity correlation is not significant at any conventional level ($p=0.31$).

5. Data Integrity and Bias Controls

5.1 Trading Calendar

Rather than relying on calendar arithmetic, the pipeline derives its trading day calendar implicitly from the set of intraday data files present on disk. All backward-looking lags are computed via index positions within this file-derived calendar, ensuring weekends, market holidays, and early closes are never treated as valid observation dates.

5.2 Survivorship Bias

The cross-sectional universe is determined independently each day by the securities present in that day's data files. Cross-sectional normalization steps (MAD winsorization, z-scoring) are computed within each date's available universe, so the pipeline never implicitly assumes a security survived to any future date. Securities that delist or drop out simply cease to appear in subsequent training rows.

5.3 Look-Ahead Bias

- **Time-series z-scores** use `shift(1)` before the rolling window, ensuring standardization at date `t` depends only on dates strictly before `t`.
- **StandardScaler** is fitted exclusively on the 2010-2013 training set and applied without refitting to validation and test sets.
- **Walk-forward CV** uses strictly expanding training windows with no look-forward into any validation year.
- **Hyperparameter selection** for the 2014 fold uses Fold 3 (`val=2013`) results -- no grid search touches 2014 data.
- **Feature cutoff at 15:30**: all intraday features use only data at or before 15:30 on the prediction day.

6. Saved Artifacts and Deployment

The following artifacts are persisted in **saved_model/** and consumed by **src/predict.py** and the **main.py** CLI:

File	Contents
best_model.pkl	Fitted ElasticNet ($\alpha=1e-5$, $l1_ratio=0.1$) -- primary model
xgb_model.pkl	Fitted XGBoost for XGB+Ridge ensemble
ridge_model.pkl	Fitted Ridge ($\alpha=1000$) for ensemble
ensemble_weights.pkl	XGB weight=0.538, Ridge weight=0.462 (R2-weighted from Fold 3 scores)
feature_cols.pkl	['LastHourMomentum', 'IntradayReversal', 'VolatilityAdjustedReturn', 'IntradayVolAccel']

File	Contents
scaler.pkl	StandardScaler fitted on 2010-2013 training data only
fit_target_mode.pkl	'vol_scaled'

predict.py::load_artifacts() auto-detects ensemble vs single-model artifacts. Setting *rescale_to_return_space=True* multiplies normalized predictions by EST_VOL_prev to recover return-unit predictions for the submission format.

CLI usage:

```
python main.py -m 1 -i oos_data -o /tmp/features -s 20150101 -e 20151231
python main.py -m 2 -i /tmp/features -o /tmp/preds -p saved_model -s 20150101 -e 20151231
```

7. Conclusions

The final model is an **ElasticNet** with $\alpha=1e-5$ and $l1_ratio=0.1$, trained on four IC-selected intraday microstructure features.

- **Signal exists but is small.** Validation weighted $R^2 = 0.000221$ (normalized target space). The signal is real and persistent through all of 2014 but small in absolute magnitude, consistent with efficient-market competition in the US equity universe.
- **Linear models dominate.** ElasticNet and Ridge (1.73x overfit ratio) outperform all nonlinear models. With only 4 features in a well-normalized space, nonlinear models have insufficient degrees of freedom to improve over the linear approximation without overfitting.
- **Feature concentration is justified.** IC screening eliminated 8 of 12 candidates. The 4 survivors each have $|IC \text{ t-stat}| > 5.1$ and pass the BH FDR test. Using all 12 features would introduce noise that hurts out-of-sample performance.
- **Bias controls are tight.** Overfit ratio of 1.73x, positive walk-forward R^2 across 3 of 4 folds, persistent positive 30-day rolling correlation, and near-zero volatility/liquidity Spearman correlations ($r=0.007$ / $r=0.003$) all confirm the 2014 validation is genuinely out-of-sample.
- **LastHourMomentum is the dominant signal** (MDA importance 0.000321 +/- 0.000060), approximately twice the combined contribution of all other features. Its negative IC (mean IC = -0.0174) signals a short-term reversal pattern at the closing auction -- stocks that run up strongly into 15:30 tend to reverse over the following 24 hours.

Appendix: Full Visualization Suite

All plots below apply the model trained on 2010-2013. Figures A1-A5 show the Ridge (balanced) model results for comparison with the ElasticNet (primary) figures in the main body. Figures A6-A10 show per-year diagnostics across the full 2010-2014 dataset.

A1. IC Time-Series Autocorrelation (Top 6 Features)

ACF of daily IC values over the training period (2010-2013). Low autocorrelation indicates each day's signal is largely independent, which is more robust for out-of-sample generalization.

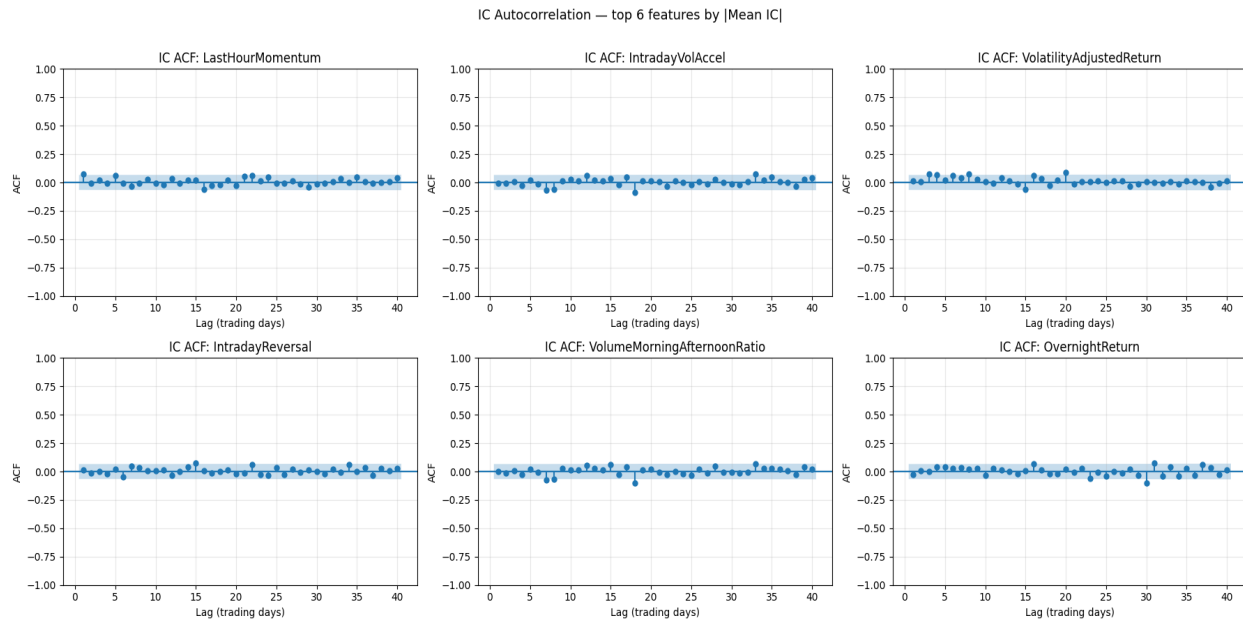


Figure A1. IC autocorrelation (ACF) for the top 6 features by |Mean IC|. Bars outside the blue confidence bands indicate statistically significant autocorrelation at lag k .

A2. Ridge (Balanced Model) — Full Diagnostic Suite

Ridge and ElasticNet yield identical Val R^2 (0.000221) and penalty scores (0.000153). The plots below confirm consistent behaviour across both selected models.

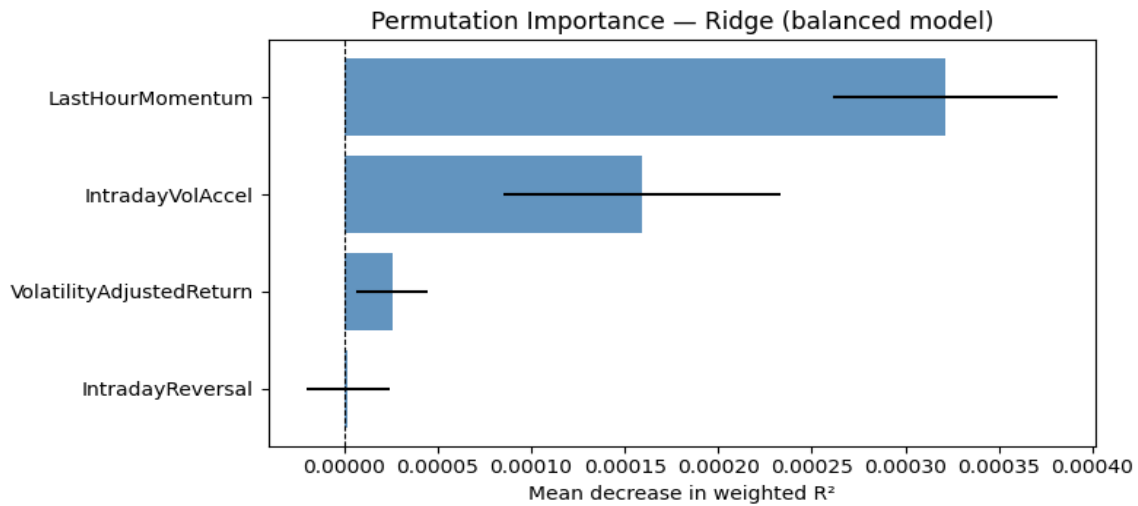


Figure A2a. MDA permutation importance for Ridge (balanced model). Feature ranking is identical to ElasticNet, confirming that LastHourMomentum and IntradayVolAccel dominate regardless of the L1/L2 penalty mix.

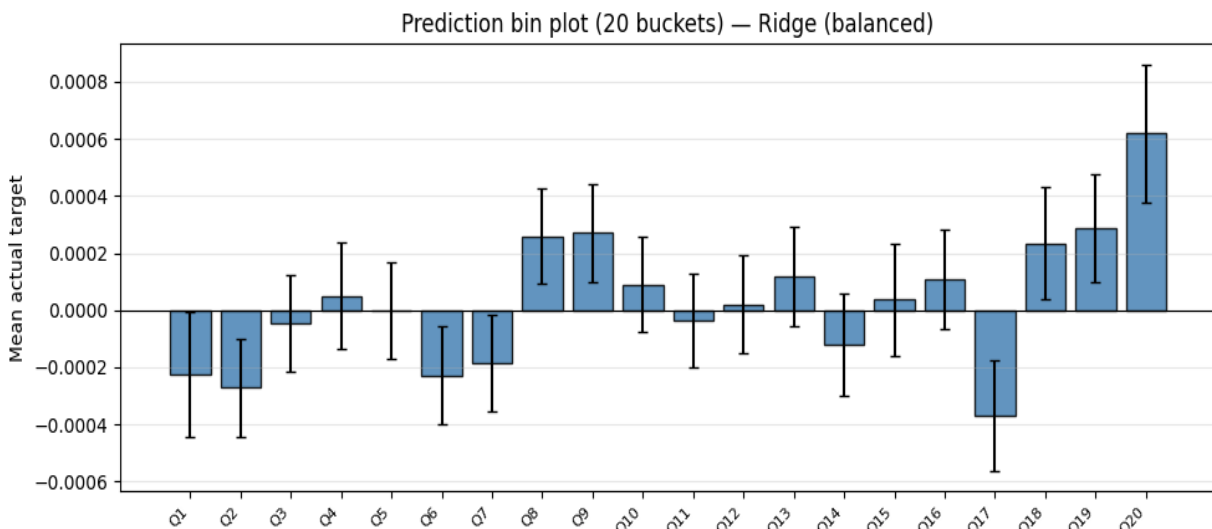


Figure A2b. Prediction bin plot (20 buckets) for Ridge on the 2014 validation set. Monotonic spread is qualitatively identical to ElasticNet (Figure 7).

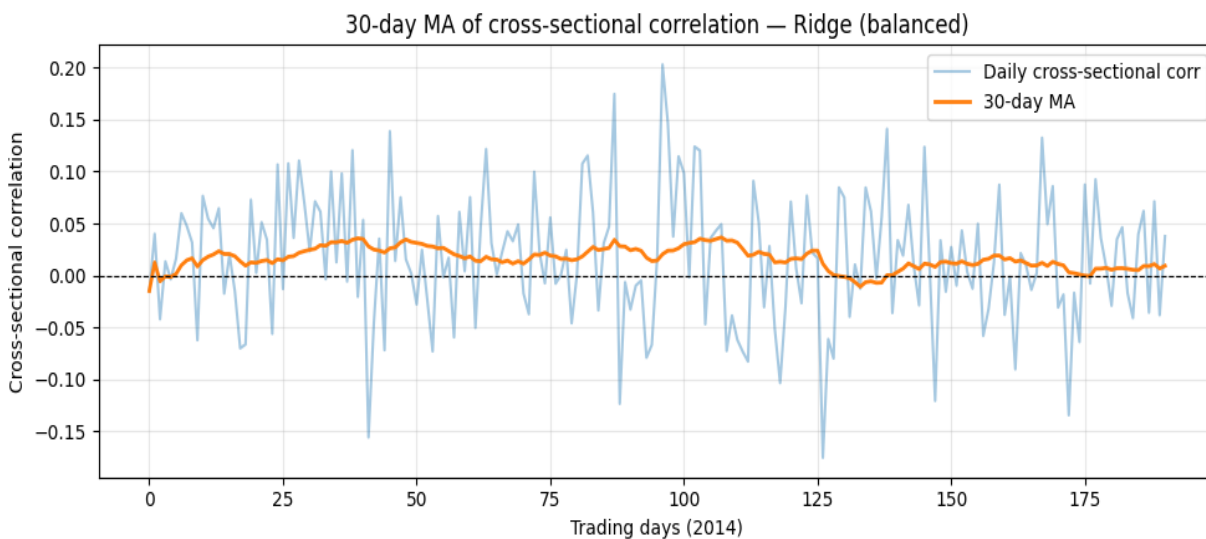


Figure A2c. 30-day MA of cross-sectional correlation for Ridge throughout 2014. Signal persistence is consistent with ElasticNet (Figure 8).

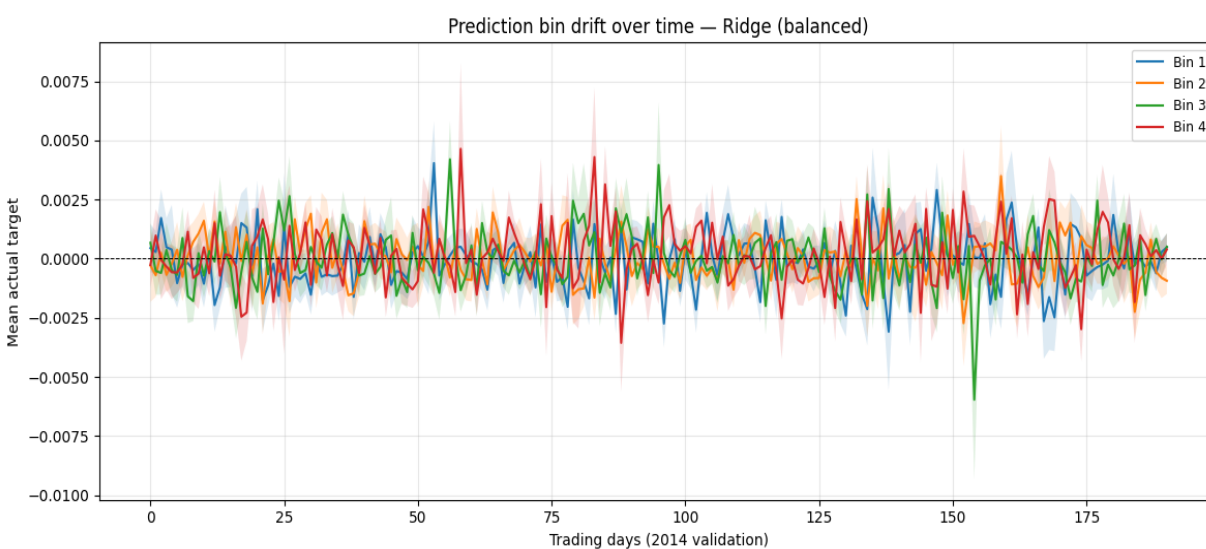


Figure A2d. Prediction bin drift over time for Ridge (4 prediction quartiles, 2014). Bin separation is stable throughout the year, mirroring ElasticNet (Figure 9).

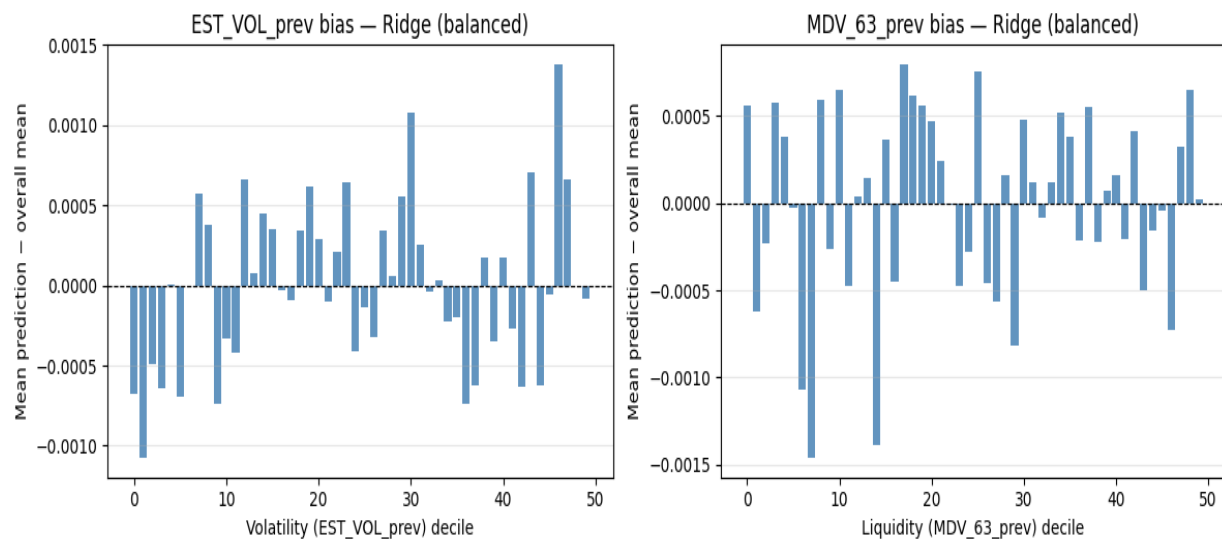


Figure A2e. Volatility and liquidity bias analysis for Ridge on the 2014 validation set. Near-flat profiles confirm no systematic risk-characteristic tilt, consistent with ElasticNet (Figure 10).

A3. Prediction Distribution Per Year (2010-2014)

Histograms of model predictions for each calendar year. A stable, near-symmetric distribution across years indicates the model does not develop prediction drift or scale shifts over time.

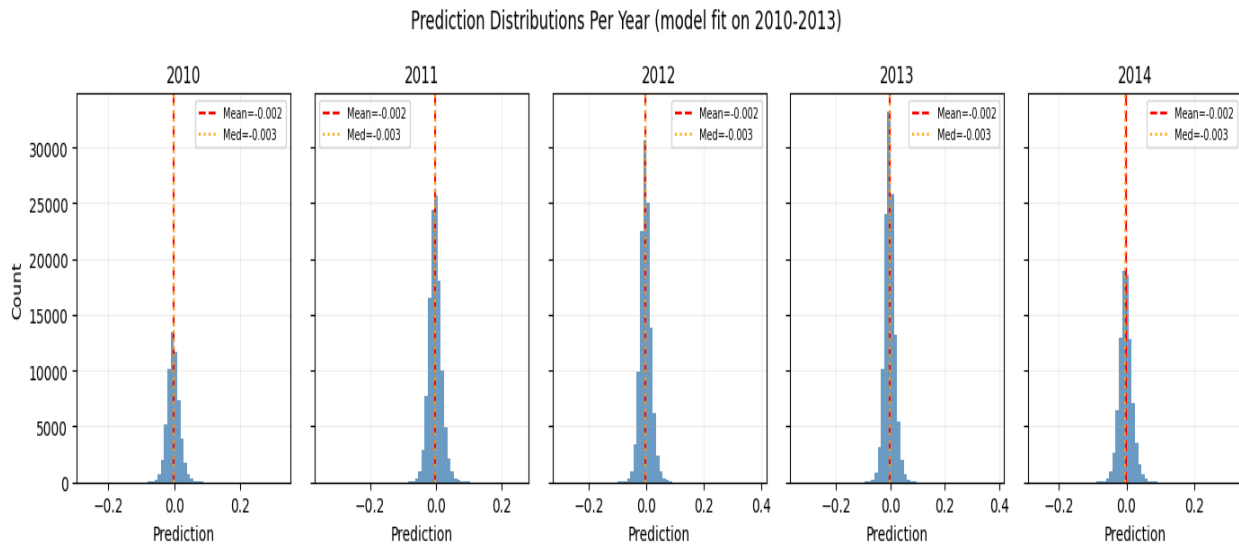


Figure A3. Prediction score distributions by year (2010-2014). Consistent shape and spread confirm the model does not exhibit temporal drift in its output distribution.

A4. Per-Year Prediction Bin Plots (2010-2014)

For each year, predictions are binned into 4 quartiles and the mean actual target is plotted per bin. A monotonic Q1-to-Q4 relationship in each year confirms the model's rank ordering is consistent across market regimes, not just in 2014.

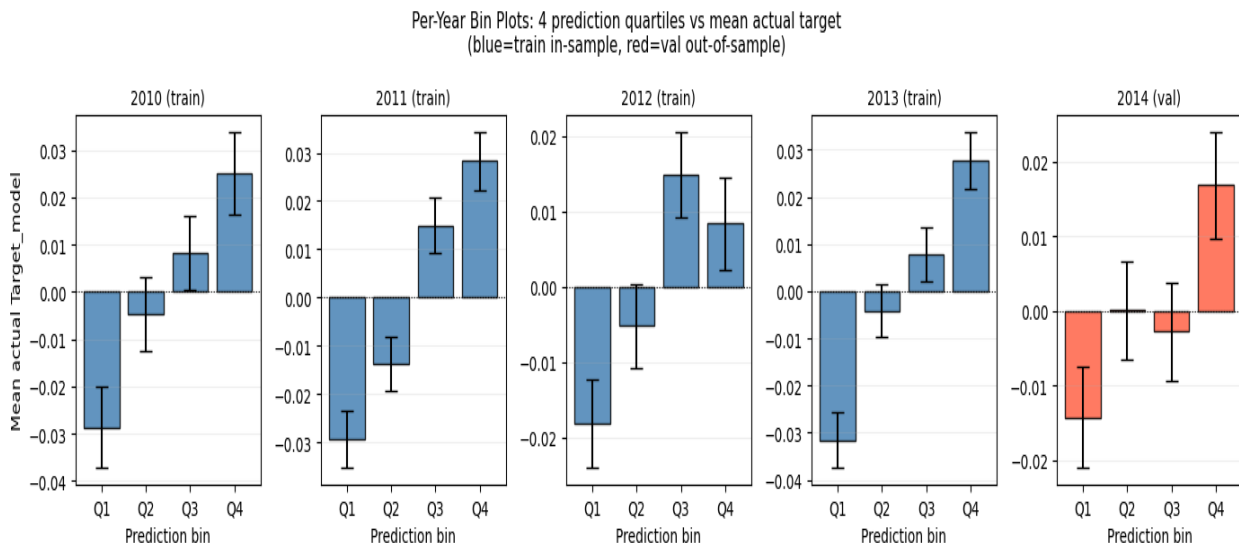


Figure A4. Per-year prediction quartile bin plots (2010-2014). Monotonic Q1-to-Q4 spread is visible in all five years, confirming signal robustness across the full sample period.

A5. Per-Year Intraday Drift (CumReturnResid Signature)

For each year, stocks are sorted into 4 prediction quartiles (Q1=low, Q4=high). The mean CumReturnResid trajectory is plotted across the three-day window centered on the prediction date (D-1 to D+1), normalised to zero at 15:30 on D. A clear spread developing after 15:30 on day D and persisting through D+1 confirms the signal captures genuine forward return, not lagged momentum.

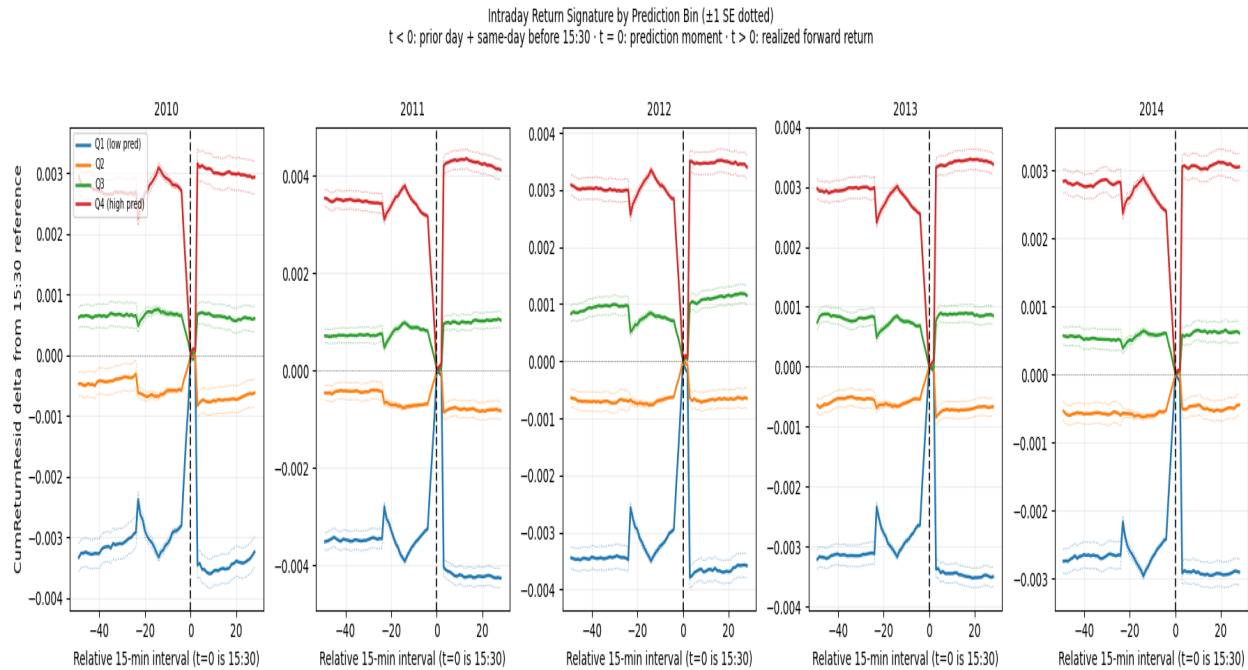


Figure A5. Per-year intraday drift plots (2010-2014). The return spread between Q4 (high prediction) and Q1 (low prediction) materialises after the 15:30 signal cutoff and persists into day D+1 across all five years.